General data processing

One of the biggest prospects of embedded systems is the processing of data and the elimination of errors. This will be done by integrating multiple processors within complex systems. So as a result, embedded technology can bring together, with maximum efficiency, the functioning of multiple cores in computer systems, hence increasing the speed and accuracy by a great margin.

Here is a quick look at the various industries and how embedded systems contribute to the specific devices:

Consumer function

- Washing Machines to control the water and spin cycles
- Remote controls that accept key touches and send infrared (IR) pulses
- Exercise equipment that measures speed, distance, calories, etc
- Clocks and watches that maintain the time, alarm and display
- Games and toys to entertain people, joystick input, video output

eCommunication

- Answering machines: Plays outgoing message, saves and organises messages
- Telephone system: Interactive switching and information retrieval
- Cellular phones: Key pad and inputs, sound I/O, and communication
- ATM machines: Provides both security and banking convenience

Automotive

- Automatic braking: Optimises stopping on slippery surfaces
- Noise cancellation Improves sound quality by removing background noise
- Theft deterrent devices Keyless entry, alarm systems
- Electronic ignition Controls sparks and fuel injectors
- Power windows and seats: Remembers preferred settings for each driver
- Instrumentation: Collects and provides the driver with necessary information

Military

- Smart weapons: Can recognise friendly targets
- Missile guidance systems: Directs ordinance at the desired target
- Global positioning systems: Determines where you are on the planet

Industrial

- Setback thermostats: Adjusts day/night thresholds, thus saving energy
- Traffic control systems: Senses car positions and controls traffic lights